Gold Price Prediction with Machine Learning Models

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Purpose & Background

- Help investors make more informed decisions on gold prices
- Evaluate and compare ML models ability capture commodity prices over time
- Forecasting gold is challenging due to unpredictable macroeconomic forces, but the potential value of accurate predictions makes it worth attempting to model

Data

Dataset:
- 578 monthly observations between Jan 1978 - Feb 2026
- Gold Price recorded in 19 currencies

Preprocessing
- Filtered from Jan 2018 due to changes in trends/patterns over the years; imputing the data before 2018 ensures more accurate predictions
- Clean commas, USD to numeric
- Compute lag features (lag 1 to lag 8)
- Log-transform USD price for Bayesian model

Methods

Time Series - ARIMA(3,1,3)
- Model order (p=3, q=3) was selected based on Sharma, R. K. (2016), who found similar ARIMA specifications performed well on monthly gold price data. d=1 was confirmed by inspecting the ACF of the differenced series.
- Equation: $\Delta y_t = \mu + \varphi_1 \Delta y_{t-1} + \varphi_2 \Delta y_{t-2} + \varphi_3 \Delta y_{t-3} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \theta_3 \varepsilon_{t-3} + \varepsilon_t$

Bayesian Regression - AR(1) MCMC

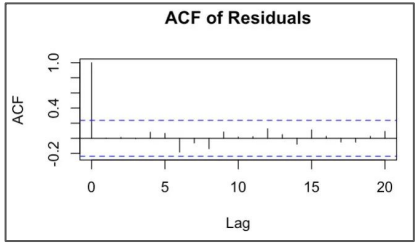
Bayesian Framework
- Place priors on parameters
- Update beliefs using observed data → posterior distributions

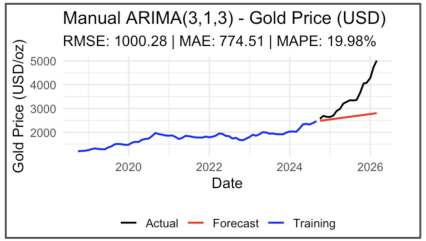
Inference via MCMC
- Sample from posterior using NUTS to estimate parameter uncertainty

Forecasting
- Simulate future prices recursively using posterior samples
- Range of future prices create uncertainty band

Results

Time Series - ARIMA(3,1,3)
- All autocorrelation values fall within the 95%confidence bands after lag 0, which confirms no patterns in the residuals, validating the ARIMA(3,1,3) selection
- The forcased red line trends upward predicting that gold would keep rising at its historical average of $18 per month but significantly underestimates the sharp growth from 2025 onward
- On average the model's predictions were $774 away from the actual price. The larger RMSE of $1,000 shows that some months were much worse than others, particularly toward the end of the 18-month window when the real price spiked sharply





Bayesian Regression - AR(1) MCMC
- Trace plots show good convergence (stable, well-mixed chains)
- Posterior distributions are smooth and unimodal
- beta_0 (drift) is small and consistent
- sigma (volatility) is stable and reasonably low
- phi (autoregression) slightly higher than 1; may be skewed as increased predictions compound
- Overall, model fit appears statistically sound
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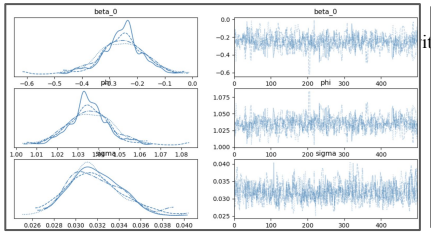

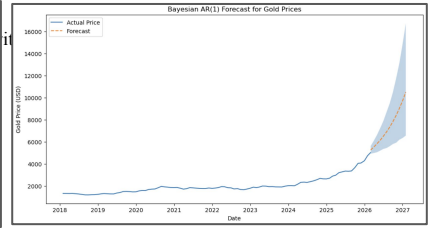

Conclusions

- ARIMA(3,1,3) captured short term trends ($\varphi_1 = 0.142$) but on performed poorly on the test dataset by ~20% on average over 18 months
- ARIMA models are worse at predicting the further you forecast
- The Bayesian model found that gold prices were strongly correlated to previous month's prices ($\varphi_1 = 0.142$)
- Bayesian also gives a full probability distribution over future outcomes that expanded wider as time goes on

Future Work

- Explore LSTM for long range and nonlinear dependencies gold prices
- Compare bayesian model accuracy with recent gold data
- Test ARIMA model with a small dataset, only a ~20 months of observations
- Add external features like inflation, interest rates, USD index, or market indicators
- Introduce time-varying volatility (e.g., stochastic volatility models) to improve uncertainty modeling

References

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